Phase 3: Python Analysis

Task 3 — Marketing Funnel & Conversion Performance Analysis

1. Introduction

The objective of this phase is to analyze e-commerce customer behavior using Python. The analysis focuses on customer interactions with products, including product views, cart additions, and purchases. The purpose is to understand user engagement, examine session-level funnel patterns, and identify useful information that can help improve marketing and conversion performance.

2. Dataset Description

The dataset is based on e-commerce behavior data from a multi-category online store. It contains event-level information about customer interactions with products.

The dataset includes the following columns:

- **event_time:** Date and time when an event occurred.
- **event_type:** Type of customer interaction, such as view, cart, or purchase.
- **product_id:** Unique identifier of a product.
- **category_id:** Identifier of the product category.
- **category_code:** Product category classification.
- **brand:** Brand associated with the product.
- **price:** Product price.
- **user_id:** Identifier of the customer.
- **user_session:** Identifier of the customer session.

For this analysis, the cleaned Excel dataset prepared during Phase 2 is used as the input.

3. Tools and Libraries Used

Python is used to process and analyze the dataset. The following libraries support the analysis:

- **Pandas:** Data manipulation, grouping, aggregation, and analysis.
- **OpenPyXL:** Reading Excel data and supporting spreadsheet input.
- **Pathlib:** Managing file paths and saving the analysis report.

4. Data Preparation

The cleaned dataset is loaded into Python using Pandas. The event timestamps are converted into a consistent datetime format, event labels are standardized, and product prices are converted into numeric values.

Text fields such as brand, category, and session identifiers are standardized by trimming unnecessary whitespace. Missing values are retained as missing rather than being replaced with guessed information.

These steps help ensure that the analysis uses consistent data types and reliable grouping fields.

5. Exploratory Data Analysis

The exploratory analysis examines the dataset from several perspectives.

Event Analysis: Counts the different types of customer interactions and calculates their percentage of total recorded events.

Customer and Session Analysis: Calculates the number of unique users, sessions, and products represented in the sample.

Price Analysis: Examines average, median, minimum, and maximum recorded product prices.

Category Analysis: Groups events by product category to compare activity, unique users, sessions, and average prices.

Brand Analysis: Examines product interaction patterns across brands and compares their recorded activity.

Daily Trend Analysis: Groups events by date and event type to examine changes in recorded activity over time.

6. Marketing Funnel Analysis

The marketing funnel analysis focuses on three main customer interactions:

1. **Product View:** A customer views a product.
2. **Add to Cart:** A customer adds a product to their cart.
3. **Purchase:** A customer completes a purchase.

Python groups the events by user session and identifies sessions containing views, cart additions, and purchases. It also calculates session-level overlaps between these event types.

Descriptive ratios, such as cart sessions divided by view sessions and purchase sessions divided by view sessions, can help summarize the observed activity. However, these ratios

should not automatically be treated as sequential conversion rates because the presence of events within the same session does not establish their order.

7. Data Quality Assessment

Data quality checks are performed to identify missing timestamps, missing event types, missing prices, missing user identifiers, missing session identifiers, non-positive prices, and exact duplicate records.

The analysis also measures missing values and distinct values for each column. These checks help identify potential data limitations that should be considered when interpreting the results.

8. Key Outputs

The Python analysis generates a text report containing:

- Dataset size and unique user, session, and product counts.
- Event counts and event distribution percentages.
- Session-level marketing funnel metrics.
- Product price statistics.
- Category-level and brand-level analysis.
- Daily event trends.
- Missing-value statistics and data quality checks.
- Scope limitations and interpretation notes.

9. Business Relevance

The analysis framework can help businesses understand how customers interact with products and where further investigation into the purchasing journey may be useful.

Event distribution can indicate which interactions dominate the observed activity. Category and brand analysis can help identify areas with higher recorded engagement. Session-level funnel metrics can provide an initial overview of customer behavior, while data quality checks help assess the reliability of the findings.

Further analysis using a larger and more representative dataset would be needed before making broad marketing or business decisions.

10. Limitations

This analysis uses a small sample of the original e-commerce dataset rather than the complete October 2019 data. Consequently, the results describe only the selected records and cannot be generalized to the entire month or all customers.

Additionally, session-level event presence does not prove that a customer completed the steps in a particular order. Missing values and duplicate records also require consideration when interpreting the results.

11. Conclusion

Phase 3 establishes a Python-based workflow for analyzing e-commerce events, examining session-level funnel patterns, comparing categories and brands, reviewing price distributions, and checking data quality.

The findings generated by the script can support the next phase of the project by helping identify which metrics and trends should be presented in a Power BI dashboard. All numerical findings should be taken from the actual Python analysis output rather than estimated or assumed values.